

List 2 - Fin Ele

1. **Cauchy-Schwarz inequality.** Consider a space V with inner product $(\cdot, \cdot)_V$ and induced norm $\|\cdot\|_V$. For any $v, w \in V$

$$|(v, w)_V| \leq \|v\|_V \|w\|_V \quad (2.29)$$

Prove this important inequality. This inequality is a particular case of the more general **Hölder's inequality**: Let $p, q \in [1, \infty]$ such that $\frac{1}{p} + \frac{1}{q} = 1$, $v \in L^p(\Omega)$ and $w \in L^q(\Omega)$, then $vw \in L^1(\Omega)$ and

$$\|vw\|_{L^1(\Omega)} \leq \|v\|_{L^p(\Omega)} \|w\|_{L^q(\Omega)} \quad (2.30)$$

$$\|v\|_V = \sqrt{(v, v)}$$

$$\begin{aligned} (v, w)_V &= (v, w - v + v)_V = (v, w - v)_V + (v, v)_V = (v, w - v)_V + \|v\|_V^2 = \\ &= (v - w + w, w - v)_V + \|v\|_V^2 = (v - w, w - v)_V + (w, w - v)_V + \|v\|_V^2 = \\ &= -\|v - w\|_V^2 + \|w\|_V^2 + \|v\|_V^2 - (w, v)_V \\ \Rightarrow (v, w)_V &= \frac{1}{2} (\|v\|_V^2 - \|v - w\|_V^2 + \|w\|_V^2) \end{aligned}$$

$$(\|v\|_V - \|w\|_V)^2 = \|v\|_V^2 - 2\|v\|_V \|w\|_V + \|w\|_V^2$$

$$\begin{aligned} \Rightarrow (v, w)_V &= \frac{1}{2} \left((\|v\|_V - \|w\|_V)^2 + 2\|v\|_V \|w\|_V - \|v - w\|_V^2 \right) \\ \Rightarrow |(v, w)_V| &= \frac{1}{2} \left((\|v\|_V - \|w\|_V)^2 - \|v - w\|_V^2 + 2\|v\|_V \|w\|_V \right) \leq \\ &\leq \frac{1}{2} \left((\|v\|_V - \|w\|_V)^2 - \|v - w\|_V^2 \right) + \|v\|_V \|w\|_V \leq \\ &\leq \|v\|_V \|w\|_V \end{aligned}$$

2. **Triangle inequality.** Using the Cauchy-Schwarz inequality, prove that for any $v, w \in V$

$$\|v + w\|_V \leq \|v\|_V + \|w\|_V \quad (2.31)$$

$$\begin{aligned} \|v + w\|_V^2 &= (v + w, v + w)_V = (v + w, v)_V + (v + w, w)_V \leq \\ &\leq \|v + w\|_V (\|v\|_V + \|w\|_V) \end{aligned}$$

$$\Rightarrow \|v + w\|_V \leq \|v\|_V + \|w\|_V$$

3. Prove that the sequence of Equation 2.12 is a Cauchy sequence.

Consider the space $C^0(\Omega)$ on $\Omega = [0, 1]$, equipped with the $L^2(\Omega)$ norm, $\|v\|_{L^2(\Omega)} = \left(\int_0^1 v^2 dx \right)^{1/2}$, and the sequence of continuous functions $\{v_n\}_{n=1}^\infty$ illustrated in Figure 2.6:

$$v_n(x) = \begin{cases} 0 & \text{if } 0 \leq x < \frac{1}{2}, \\ (x - \frac{1}{2})^{\frac{1}{n}} & \text{if } \frac{1}{2} \leq x \leq 1, \end{cases} \quad (2.12)$$

$$\|v_n - v_m\|_{L^2(\Omega)} = \left(\int_0^1 (v_n(x) - v_m(x))^2 dx \right)^{1/2} = \left(\int_{1/2}^1 \left((x - \frac{1}{2})^{\frac{1}{n}} - (x - \frac{1}{2})^{\frac{1}{m}} \right)^2 dx \right)^{1/2}$$

$$\left(\int_{1/2}^1 \left(x - \frac{1}{2} \right)^{\frac{2}{n}} - 2 \left(x - \frac{1}{2} \right)^{\frac{1}{n} + \frac{1}{m}} + \left(x - \frac{1}{2} \right)^{\frac{2}{m}} dx \right)^{1/2} = (1)$$

Note that

$$\int_{1/2}^1 \left(x - \frac{1}{2} \right)^k dx = \int_0^{1/2} u^k du = \frac{u^{k+1}}{(k+1)} \Big|_0^{1/2} = \frac{1}{2^{k+1}(k+1)}$$

$$\begin{aligned} u &= x - \frac{1}{2} \\ du &= dx \end{aligned}$$

$$\Rightarrow (1) = \frac{1}{2^{\frac{2}{n}+1} \left(\frac{2}{n} + 1 \right)} - \frac{2}{2^{\frac{1}{n} + \frac{1}{m} + 1} \left(1 + \frac{1}{n} + \frac{1}{m} \right)} + \frac{1}{2^{\frac{2}{m}+1} \left(\frac{2}{m} + 1 \right)} =$$

$$\frac{1}{2^{\frac{2}{n}+1} \left(\frac{2}{n} + 1 \right)} + \frac{1}{2^{\frac{2}{m}+1} \left(\frac{2}{m} + 1 \right)} - \frac{1}{2^{\frac{1}{n} + \frac{1}{m} + 1} \left(1 + \frac{1}{n} + \frac{1}{m} \right)} = (2)$$

(Pesso duridoso!!!)

Noting that (2) is continuous as a function of m and n for $m, n \geq 1$, if we take $m, n \rightarrow \infty$

$$(2) \rightarrow \frac{1}{2} + \frac{1}{2} - 1 = 0$$

As the limit exists, $\forall \epsilon > 0$, $\exists N > 0$ such that $n, m > N \Rightarrow$

$$\|v_n - v_m\|_{L^2(\Omega)} < \epsilon$$

5. Let $f: \Omega \rightarrow \mathbb{R}$, $\Omega \subset \mathbb{R}^3$. Write the partial derivative of f corresponding to $\vec{\alpha} = (2, 0, 0)$ and $\vec{\alpha} = (0, 1, 1)$.

$$\vec{\alpha} = (2, 0, 0):$$

$$\vec{\alpha} = (0, 1, 1):$$

$$D^{\vec{\alpha}} f = \frac{\partial^2 f}{\partial x_1^2}$$

$$D^{\vec{\alpha}} f = \frac{\partial^2 f}{\partial x_2 \partial x_3}$$

10. Let Ω be bounded and let $1 \leq p \leq q < \infty$. Prove the inclusion stated without proof in Section 2.3,

$$L^q(\Omega) \subset L^p(\Omega).$$

a) Apply Hölder's inequality, Eq. (2.29), to the product $|f|^p \cdot 1$ with the exponents $r = q/p$ and $r' = r/(r-1)$, and conclude that

$$\|f\|_{L^p(\Omega)} \leq |\Omega|^{\frac{1}{p} - \frac{1}{q}} \|f\|_{L^q(\Omega)},$$

where $|\Omega|$ denotes the measure of Ω . Observe where the boundedness of Ω was used.

b) Show that the inclusion is strict: take $\Omega = (0, 1)$, $p = 1$, $q = 2$ and $f(x) = x^{-s}$, $s > 0$. Determine for which s one has $f \in L^1(\Omega)$ and for which $f \in L^2(\Omega)$.

c) Show that boundedness of Ω is essential: taking $\Omega = (1, \infty)$ and $f(x) = 1/x$, verify that $f \in L^2(\Omega)$ but $f \notin L^1(\Omega)$.

$$L^p(\Omega) = \left\{ f: \Omega \rightarrow \mathbb{R} \mid \int_{\Omega} |f|^p dx < +\infty \right\}$$

In fact,

$$\|f\|_{L^p(\Omega)} = \left(\int_{\Omega} |f|^p dx \right)^{1/p}$$

$$\begin{aligned} \frac{1}{r} + \frac{1}{r'} &= \frac{1}{r} + \frac{r-1}{r} \\ &= \frac{r}{r} = 1 \end{aligned}$$

$$\|f\|_{L^p(\Omega)} = \left(\int_{\Omega} |f|^p dx \right)^{1/p} = \|f\|_{L^p(\Omega)}^p \quad (1)$$

we can apply Hölder inequality for these factors

And

$$\|f\|_{L^p(\Omega)} = \| |f|^p \|_{L^1(\Omega)}^{1/p} \stackrel{\text{Hölder}}{\leq} \| |f|^p \|_{L^r(\Omega)}^{1/p} \|1\|_{L^{r'}(\Omega)}^{1/p} =$$

$$= \left(\int_{\Omega} |f|^p dx \right)^{\frac{p}{p}} \left(\int_{\Omega} 1^{r'} dx \right)^{\frac{1}{p}} = \left(\int_{\Omega} |f|^p dx \right)^{\frac{p}{p}} \left(\int_{\Omega} 1 dx \right)^{\frac{1}{p}} =$$

$$= |\Omega|^{\frac{1}{p}} \left(\int_{\Omega} |f|^p dx \right)^{\frac{p}{p}}$$

$$\frac{1}{r'} = \left(\frac{p}{p} - 1 \right) \frac{p}{p} = \frac{p-p}{p} = \frac{p-p}{p} \Rightarrow \| |f|^p \|_{L^1(\Omega)} = |\Omega|^{\frac{p-p}{p}} \|f\|_{L^p(\Omega)}^p$$

Using (1):

$$\|f\|_{L^p(\Omega)} = \| |f|^p \|_{L^1(\Omega)}^{1/p} = |\Omega|^{\frac{p-p}{p}} \|f\|_{L^p(\Omega)}^p = |\Omega|^{\frac{1}{p} - \frac{1}{p}} \|f\|_{L^p(\Omega)}^p$$

If Ω is not bounded, then $L^p(\Omega)$ is not finite, which gives us nothing useful, as $\|f\|_{L^p(\Omega)}$ could be ∞ .

This shows that if $f \in L^p(\Omega)$, then $\|f\|_{L^p(\Omega)} < \infty \Rightarrow f \in L^p(\Omega)$.

b) $\Omega = (0,1)$; $p=1$; $q=2$
 $f(x) = x^{-s}$, $s > 0$

1) Finding s such that

$$f \in L^1(\Omega) \Leftrightarrow \int_{\Omega} |f(x)| dx < \infty$$

$$\int_{\Omega} |f(x)| dx = \int_0^1 x^{-s} dx$$

if $s < 1$ $\int_{\Omega} |f(x)| dx = \frac{x^{1-s}}{1-s} \Big|_0^1 = \frac{1^{1-s}}{1-s} - \frac{0^{1-s}}{1-s} = \frac{1}{1-s} < \infty$

if $s > 1$ $\int_{\Omega} |f(x)| dx = \frac{1}{(1-s)x^{s-1}} \Big|_0^1 = \frac{1}{1-s} - \lim_{x \rightarrow 0^+} \frac{1}{(1-s)x^{s-1}} \rightarrow \infty$

if $s = 1$ $\int_{\Omega} |f(x)| dx = \ln(x) \Big|_0^1 = 0 - \lim_{x \rightarrow 0^+} \ln(x) \rightarrow \infty$

$f \in L^1(\Omega)$ when $s < 1$

2) Finding s such that $f \in L^2(\Omega)$

$$f \in L^2(\Omega) \Leftrightarrow \int_{\Omega} |f(x)|^2 dx < \infty$$

$$\int_{\Omega} |f(x)|^2 dx = \int_{\Omega} x^{-2s} dx = \int_{\Omega} x^{-t} dx, \text{ with } t = 2s.$$

We know this converges for $t < 1 \Rightarrow s < \frac{1}{2}$

So, $\forall s \in [\frac{1}{2}, 1)$, $f \in L^1(\Omega)$ but $f \notin L^2(\Omega)$

$\Rightarrow$ For $1 \leq p < \infty$, $\exists f \in L^p(\Omega)$ such that $f \notin L^q(\Omega) \Rightarrow L^p(\Omega) \not\subset L^q(\Omega) \Rightarrow L^q(\Omega) \subsetneq L^p(\Omega)$

c) $\Omega = (1, \infty)$ $f(x) = \frac{1}{x}$

$$\int_{\Omega} |f(x)|^p dx = \int_1^{\infty} \frac{1}{x^p} dx = \frac{-1}{x^{p-1}} \Big|_1^{\infty} = -\lim_{x \rightarrow \infty} \frac{1}{x^{p-1}} - \left(-\frac{1}{1}\right) = 0 + 1 = 1 < \infty$$

$\Rightarrow f \in L^p(\Omega)$

$$\int_{\Omega} |f(x)| dx = \int_1^{\infty} \frac{1}{x} dx = \ln(x) \Big|_1^{\infty} = \lim_{x \rightarrow \infty} \ln(x) = \infty$$

$\Rightarrow f \notin L^1(\Omega)$

$\Rightarrow$ The boundedness of Ω is necessary for the inclusion to be valid.

13. Consider again the function $f(x) = 1 - |x|$ on $\Omega = (-1, 1)$ introduced earlier, whose first weak derivative is the piecewise constant function

$$g(x) = \begin{cases} 1, & -1 < x < 0, \\ -1, & 0 < x < 1, \end{cases}$$

We wish to show that the second weak derivative of f does not exist.

a) Suppose that $h = D^2 f$ exists. Repeating on g the integration by parts performed in earlier, splitting the interval at $x = 0$ and recalling that g is constant on each half, show that necessarily

$$\int_{-1}^1 h(x) \varphi(x) dx = -2 \varphi(0) \quad \forall \varphi \in \mathcal{D}(\Omega).$$

Note that, unlike the previous case, the boundary terms at $x = 0$ no longer cancel, because g is discontinuous there. Identify the functional on the right-hand side as $-2\delta_0$, where δ_0 is the Dirac delta, and explain informally why no ordinary function can reproduce it.

b) (Optional for undergrads) Prove it rigorously. Let $\varphi_1 \in \mathcal{D}(-1, 1)$ satisfy $\varphi_1(0) = 1$ and $0 \leq \varphi_1 \leq 1$ (for instance the bump function of Figure 2.8), and set $\varphi_n(x) = \varphi_1(nx)$.

- Verify that $\varphi_n \in \mathcal{D}(\Omega)$ and that $\text{supp } \varphi_n \subset (-1/n, 1/n)$, so that the identity of part (a) gives $\int_{-1}^1 h \varphi_n dx = -2$ for every n .
- Show that, for any $h \in L^1_{\text{loc}}(\Omega)$,

$$\left| \int_{-1}^1 h \varphi_n dx \right| \leq \int_{-1/n}^{1/n} |h| dx \xrightarrow{n \rightarrow \infty} 0,$$

justifying the limit by the absolute continuity of the Lebesgue integral. Conclude that $-2 = 0$, a contradiction, so no such h exists.

Being continuous and piecewise C^1 therefore buys one weak derivative, but not two. Continuity by itself buys nothing: a Weierstrass function is continuous and possesses no weak derivative at all. This is why continuous piecewise linear finite elements belong to $H^1(\Omega)$ but not to $H^2(\Omega)$.

a) Let $\varphi \in C^\infty(\Omega)$ such that $\lim_{x \rightarrow 1^-} \varphi(x) = \lim_{x \rightarrow -1^+} \varphi(x) = 0$

$$\int_{-1}^1 g(x) \varphi'(x) dx = \int_{-1}^0 \varphi'(x) dx - \int_0^1 \varphi'(x) dx = \varphi(x) \Big|_{-1}^0 - \varphi(x) \Big|_0^1 = \varphi(0) - (-\varphi(0)) = 2\varphi(0)$$

Supposing h is a weak derivative for g , we have

$$-\int_{-1}^1 h(x) \varphi(x) dx = \int_{-1}^1 g(x) \varphi'(x) dx = 2\varphi(0) \Rightarrow$$

$$\Rightarrow \int_{-1}^1 h(x) \varphi(x) dx = -2\varphi(0) = -2 \int_{-1}^1 \delta_0(x) \varphi(x) dx$$

As g is differentiable at $x \neq 0$, we must have $h = g'$ for $x \neq 0$. But then, $h(x) \neq 0$ only at $x = 0$. But $\{0\}$ is a set with measure zero. If $h(0)$ is finite, then we'd have $\int_{-1}^1 h(x) \varphi(x) dx = 0$, $\forall \varphi \in \mathcal{D}$, absurd. So, h must be a Dirac's delta.

b) $\varphi_i \in \mathcal{D}(-1, 1) = \{\varphi \in C^\infty(-1, 1) \mid \text{supp } \varphi \text{ is compact in } (-1, 1)\}$
 $\text{supp } \varphi_i = \{x \in \Omega : \varphi_i(x) \neq 0\}$

$$\varphi_i(0) = 1; \quad 0 \leq \varphi_i \leq 1$$

$$\varphi_n(x) = \varphi_i(nx)$$

As φ_i is defined only at $(-1, 1)$, $\varphi_n(x) = \varphi_i(nx) \Rightarrow \varphi_n$ is defined at $(-\frac{1}{n}, \frac{1}{n}) = \Omega_n$